G4 HIRING CHALLENGE · OVERVIEW

AI Master Challenge

Four business problems, what each one is really testing, and how to beat the AI baseline

Source: github.com/codessentia/ai-master-challenge

HOW IT WORKS

One challenge, 4–6 hours, one pull request

01

Fork and pick one

Choose a single challenge. One done well beats two done shallowly.

02

Solve with any AI

Claude, GPT, Cursor, Claude Code — any tool is allowed and expected.

03

Log the process

Prompts, iterations, where AI was wrong. No process log means disqualified.

04

Open a PR

Branch submission/your-name, folder submissions/your-name/, template README.

⚠ **The baseline:** G4 already ran every brief through Claude, GPT and Gemini. A submission that looks like that output adds nothing.

WHAT IS GRADED

Judgment is the product, not the output

Earns points

- Problem understood before any tool is opened
- Numbers that can be checked, causation kept apart from correlation
- Actions a non-technical executive can take on Monday
- Knowing what **not** to automate
- A log that shows iteration and corrections

Gets rejected

- Generic output that fits any company
- "The AI said so" with no verification
- One prompt, one answer, submit
- Looking clever instead of solving the problem
- 40 pages where 5 would do

CHALLENGE 001 · DATA / ANALYTICS · THE PROBLEM

Churn diagnosis at RavenStack

> *"Churn is up, CS says satisfaction is fine, Product says usage grew. Something doesn't add up."*

The CEO of a B2B SaaS with ~500 accounts (paraphrased)

The ask

The root cause, the segments and named accounts at risk, and prioritized actions with estimated impact.

The data: 5 linked tables

Accounts (~500) · subscriptions (~5K) · daily feature usage (~25K) · support tickets (~2K) · churn events with reasons and feedback (~600)

The hidden trap

Every team's metric can be true on average and wrong for the segments that leave. Looking at one table misses the point.

CHALLENGE 001 · POSSIBLE SOLUTION

Reconcile the three stories, account by account

1 · Analyse

- Join all 5 tables into one row per account
- Churned vs retained: usage trend, errors, tickets, CSAT
- Split by plan, industry, channel, billing; weight by MRR lost

2 · Build

- Risk score per active account, with its reasons
- Ranked list of named accounts and MRR at risk
- Validated on held-out churn events

3 · Stand out

- Test "usage grew" per segment, not in aggregate
- LLM-tag churn feedback text; compare stated reasons with behaviour
- Say which findings are causal and what test would prove it

Watch out: count revenue, not logos. Losing a $50/month account is not losing a $5K/month one.

CHALLENGE 002 · OPERATIONS / CX · THE PROBLEM

Support redesign for ~30K tickets a year

> *"Show me where we lose time, what AI can automate, and that it works. I don't want a PowerPoint."*

The Director of Operations (paraphrased)

The ask

A bottleneck diagnosis in hours and cost, an automation plan that says what to leave to humans, and a working prototype.

The data: 2 datasets

~30K customer tickets with channel, priority, type, response and resolution times, CSAT and full text · ~48K IT tickets labelled in 8 topic groups

The hidden trap

Automating 100% is a red flag. The value is in crossing both datasets and drawing a clear line where humans stay.

CHALLENGE 002 · POSSIBLE SOLUTION

A triage copilot with a human lane

1 · Analyse

- Resolution time by channel × priority × type
- Drivers of CSAT with effect sizes, or proof there are none
- Wasted hours × loaded hourly cost

2 · Build

- Classifier on the 48K IT tickets: category + confidence
- Low confidence goes to a human queue
- Suggested replies from similar resolved tickets

3 · Stand out

- App: paste a ticket, get route, priority and a draft reply
- "Never automate" list: disputes, critical, angry customers
- ROI: hours per month saved at a realistic auto-route rate

Watch out: report accuracy on thousands of held-out tickets, not three cherry-picked demos.

CHALLENGE 003 · SALES / REVOPS · THE PROBLEM

A lead scorer reps will actually open

> *"Reps waste time on deals that won't close and let good ones go cold. I want a tool, not a notebook."*

The Head of Revenue Operations (paraphrased)

The ask

Working software on real data that prioritizes deals and tells the rep **why** each one scores high or low.

The data: a 4-table CRM

Accounts (~85) · products (7) · sales teams (35 reps with managers and regions) · pipeline (~8.8K opportunities with stage, dates and value)

The hidden trap

Sorting by deal value isn't scoring, and a score without a reason gets ignored. It also has to run from the README.

CHALLENGE 003 · POSSIBLE SOLUTION

A Monday-morning focus list

1 · Analyse

- Win rates from closed deals by product, sector, account size, rep
- Deal age vs win odds: when does a deal go cold?
- Data quality: name mismatches, missing dates

2 · Build

- Score = win odds × value, discounted for staleness
- Web app filtered by rep, manager and region
- Top 3 reasons per deal, in plain words

3 · Stand out

- Buckets: act today, at risk, let go
- Back-test: do high scores really win more often?
- Commit history doubles as the process log

Watch out: the data ends years ago. Measure deal age against the dataset's last date, not today's clock.

CHALLENGE 004 · MARKETING · THE PROBLEM

Social media strategy from 52K posts

> *"What really drives engagement, is sponsoring influencers worth it, and what should we post?"*

The Head of Marketing (paraphrased)

The ask

A data-backed analysis, a sponsorship policy, what to stop doing, and quick wins for this week.

The data: one wide table

~52K posts on YouTube, TikTok, Instagram, Bilibili and RedNote · content type, category, hashtags · views, likes, shares · followers, audience demographics · sponsorship flag

The hidden trap

A raw sponsored vs organic average is unfair, and "video beats image" is the baseline answer.

CHALLENGE 004 · POSSIBLE SOLUTION

Fair comparisons, then a short playbook

1 · Analyse

- Sponsored vs organic within creator size × platform × category
- Engagement relative to the platform median
- What fails, including zero-engagement posts

2 · Build

- Sponsorship policy with follower and engagement thresholds
- Ranked do / stop / quick-win list
- Hashtag combinations linked to lift

3 · Stand out

- Predictor: post features in, expected engagement range out
- Small weekly dashboard for the social team
- One-page brief readable in five minutes

Watch out: survivorship bias. Top posts are visible; the long tail of flops is where the signal is.

THE EDGE ACROSS ALL FOUR

Distrust the data before you analyse it

These are public Kaggle datasets, and some may be generated. Pasting the brief into an AI skips this step. Doing it is your clearest proof of judgment.

Check the shape

Are distributions realistic, or suspiciously uniform?

Check the signal

Does the target actually vary with anything?

Say it plainly

No signal? Report it with evidence. Don't invent a story.

SIDE BY SIDE

Which challenge fits which profile

Challenge	Best fit	Core deliverable	Hardest part	Build effort
001 Churn	Data / analytics	Diagnostic report	Root cause, not correlation	Medium
002 Support	Ops + ML	Diagnosis + classifier	Most to deliver in 4–6 h	High
003 Lead scorer	Software builder	Working app (required)	Explainable scores	High
004 Social	Marketing analytics	Strategy + policy	Fair sponsorship comparison	Low–medium

RECOMMENDATION

Pick 003 if you build, 001 if you analyse

003 turns Claude Code's commit history into a ready-made process log. 001 has the richest analytical trap, built for someone who joins data well.

Next steps

1. Choose the challenge
2. Fork, create submission/your-name, download the data
3. Audit the data before analysing it
4. Build and commit in small steps
5. Write the README and the process log
6. Open the pull request